

The circuit turns the mosquito repellent periodically 'on' and 'off' for approximately 20 minutes each. So if you leave the mosquito repellent switched on from 10 pm to 6 am (eight hours), it will be 'on' for four hours and 'off' for four hours of the total duration. During 'off' time, the room air tries to balance its natural composition. Another important feature is that the circuit switches to 'on/off'

operation without producing any noise or a sound click as in a relay and hence doesn't disturb your sleep.

The circuit consists of a timer section built around IC 555 (IC1) and an automatic switching section using triac BT136 (TRIAC1). Power supply to the circuit is derived from the AC mains by stepping it down to a required level and rectifying it. The elimination of the transformer saves on space as well as

money. Zener diode ZD1 and capacitor C2 provide regulated 9V DC power supply to timer IC1.

The timer section comprises resistors R1 and R2 and capacitor C1. The output of timer IC 555 is fed to the gate terminal of BT136 through series LED1 and resistor R4. When the timer

output goes high, it triggers the gate of TRIAC1 and LED1 shows the 'on' period. During 'off' time, the output of IC1 goes low and hence TRIAC1 is not fired and LED1 doesn't glow.

The circuit is very compact and can be assembled on a general-purpose PCB. Use an 8-pin IC base for timer IC LM555. After assembly, fit the unit inside the housing electric board where you plug in the mosquito repellent. ●

PLAY WITH ROBOTIC EYE (IR SENSOR)

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There are various applications of IR sensors such as TV remote controllers, burglar alarms and object counters. Here we have used IR sensors (infrared LEDs) to make an

object-detection circuit and also a proximity sensor for path-tracking robots. The basic idea is to transmit the infrared light through an IR LED, which is then reflected by any obstacle and sensed by the receiving LED.

On receiving the light, the sensing LED develops a voltage difference across its leads. However, this voltage produces very weak electrical signal. Therefore we have used an operational amplifier (op-amp) for accurate detection of the object.

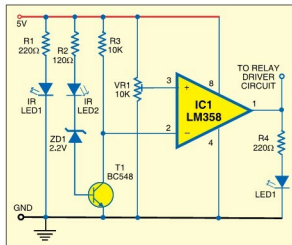


Fig. 1: Object detection circuit

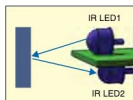


Fig. 2: Fitting of IR sensors



Fig. 1 shows the object-detection circuit. The circuit can be divided into two parts: emitter and receiver. The emitter is composed of IR LED1 in series with a 220-ohm resistor that limits the forward current. IR LED1 emits IR light continuously. The receiver is built around IC LM358 (IC1), IR LED2, Zener diode ZD1, transistor T1 and resistors R2 and R3 along with preset VR1.

According to functioning of the op-amp, the output of IC1 will go high when the voltage at its pin 2 goes high. So the output of IC1 goes high when IR light is detected by the IR receiver. When the IR light falls on IR LED2, transistor T1 conducts making pin 2 of IC1 low. You can adjust the reference voltage at pin 3 of IC1 using preset